Wazuh SIEM Deployment and Security Event Triage Lab Report

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Executive Summary

This report documents the setup, configuration, and testing of a local Security Information and Event Management (SIEM) lab using Wazuh (v4.9.2) deployed via Docker Compose on a Windows laptop running WSL 2 (CachyOS). The goal of this project was to gain hands-on Security Operations Center (SOC) experience, set up endpoint log collection, and practice Tier 1 threat triage on simulated Windows security events.

Lab Architecture & Environment

- **Host Operating System:** Windows 11 Pro with WSL 2 (CachyOS Linux distribution)
- **SIEM Platform:** Wazuh v4.9.2 (Single-node deployment consisting of `wazuh.indexer`, `wazuh.manager`, and `wazuh.dashboard`)
- **Monitored Endpoint:** Local Windows 11 host registered as `Windows-Host-01`

Implementation Steps & Evidence

1. Infrastructure Deployment

- Configured WSL 2 environment memory parameters (`vm.max_map_count=262144`) so the Wazuh indexer could run smoothly.
- Cloned the official Wazuh Docker repository (v4.9.2) and generated self-signed SSL certificates for secure internal communication.
- Launched the multi-container stack via Docker Compose and verified healthy status across all three services using `docker ps`.

```
~/wazuh-docker/single-node tags/v4.9.2 40m 44s
❯ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED       STATUS       PORTS                                                
```

- Accessed the web-based management console at `https://localhost` using administrative credentials.

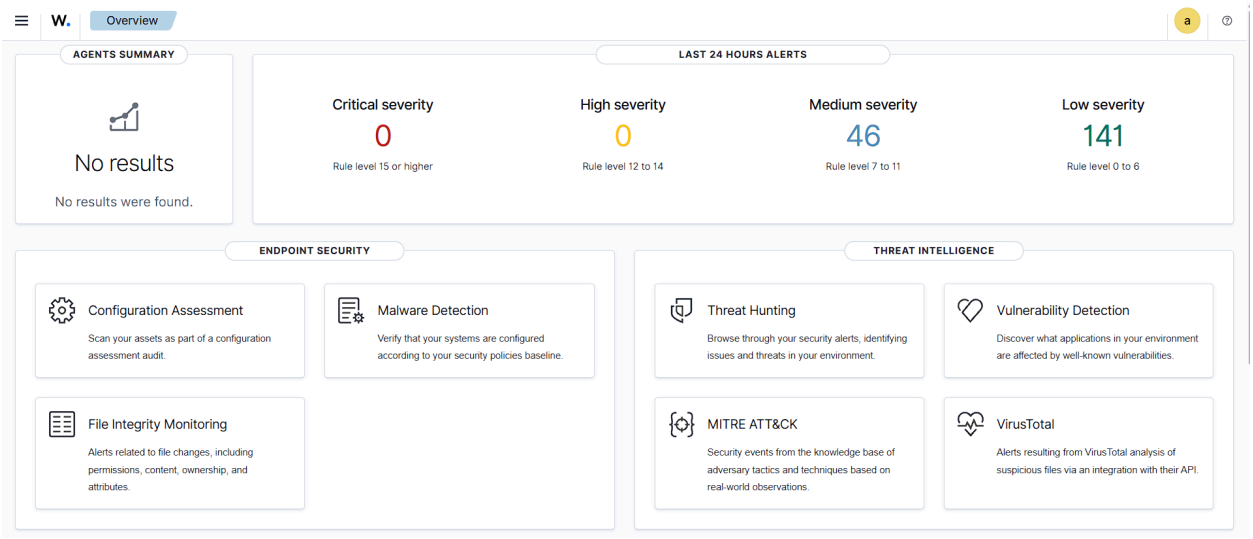


Figure 2: Wazuh Web Console Overview dashboard initialized locally.

2. Endpoint Agent Enrollment

- Deployed the Wazuh Windows agent (v4.9.2) on the host machine via Windows PowerShell using an automated MSI installation command pointing to the local loopback manager address (`127.0.0.1`).
- Started the `WazuhSvc` Windows background service to begin streaming system logs to the SIEM.
- Confirmed successful agent connection in the Wazuh dashboard, where `Windows-Host-01` achieved an active status.



Figure 3: `Windows-Host-01` enrolled and displaying active status in Endpoints Summary.

3. Attack Simulation & Threat Detection

- Performed a controlled failed login simulation on the Windows endpoint to test if Wazuh would detect suspicious authentication activity.
- Generated Windows Security Event ID `4625` (Logon Failure) by running a test network login command against the local machine using fake credentials (`FakeAttacker`).
- Verified that Wazuh ingested and parsed the security logs in the Threat Hunting dashboard, triggering Rule ID `60122` (*Logon Failure - Unknown user or bad password*) mapped to MITRE ATT&CK technique **T1110 (Brute Force)**.

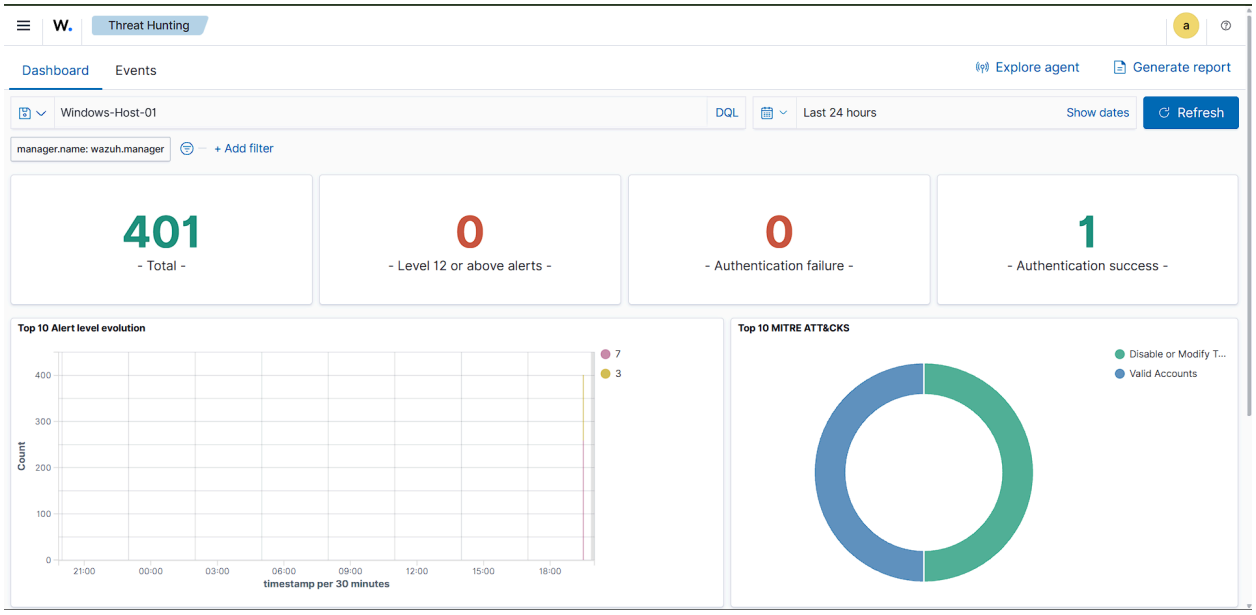


Figure 4: Wazuh Threat Hunting dashboard displaying alert timeline and MITRE ATT&CK mappings.

Key Findings & Tier 1 SOC Log Triage

During incident review in the Wazuh Threat Hunting Events view, inspecting the raw JSON event details confirmed all key forensic fields were captured:

- **Agent Name:** `Windows-Host-01` (ID: `001`)
- **Event ID:** `4625` (Windows Security Channel)
- **Rule Triggered:** Rule `60122` — *Logon Failure - Unknown user or bad password*
- **Target User:** `FakeAttacker`
- **Logon Type:** `3` (Network Logon)
- **Sub-status Code:** `0xc0000064` (User name does not exist)

```
"agent": {
  "ip": "127.0.0.1",
  "name": "Windows-Host-01",
  "id": "001"
},
"manager": {
  "name": "wazuh.manager"
},
"data": {
  "win": {
    "eventdata": {
      "subjectLogonId": "0x0",
      "ipAddress": "127.0.0.1",
      "authenticationPackageName": "NTLM",
      "workstationName": "LENOVO1GAME",
      "subStatus": "0xc0000064",
      "logonProcessName": "NtLmSsp",
      "targetUserName": "FakeAttacker",
      "keyLength": "0",
      "subjectUserSid": "S-1-0-0",
      "processId": "0x0",
      "ipPort": "52280",
      "failureReason": "%%2313",
      "targetUserSid": "S-1-0-0",
      "logonType": "3",
      "status": "0xc000006d"
    },
    "system": {
      "eventID": "4625",
      "keywords": "0x8010000000000000",
      "providerGuid": "{54849625-5478-4994-a5ba-3e3b0328c30d}",
      "level": "0",
      "channel": "Security",
      "opcode": "0",
```


Figure 5: Raw JSON document inspection showing Event ID 4625 and targetUserName `FakeAttacker`.

SOC Analyst Triage Verdict & Next Steps

- **Alert Verdict: True Positive (Simulated Attack)** — confirmed an unauthorized network logon attempt (`Logon Type 3`) using a non-existent account (`FakeAttacker`).
- **Recommended Next Steps:** In a live SOC environment, a Tier 1 analyst would verify whether the source IP address belongs to an authorized administrator, check if the same account was targeted across other endpoints, and block malicious external IP addresses at the firewall if brute-force behavior continues.

Conclusion

The Wazuh SIEM lab environment is fully operational and capable of collecting real-time endpoint telemetry, detecting unauthorized authentication attempts, and presenting structured

JSON logs for Tier 1 SOC analysis. This project validates foundational skills in log ingestion, SIEM monitoring, and incident triage workflows.